

Trevor W. Exley

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Personal Statement

Soft robotics researcher exploring the monolithic domain: systems in which structure, actuation, and sensing are co-designed and fabricated together in a single material and a single print. I leverage parametric design frameworks to pursue immediate functionality off the printed, eliminating the post-processing that stands between a finished print and a functioning robot.

Education

Ph.D., Biomedical Engineering, University of North Texas 2021–2024
Dissertation: “Thermo-Reversible Phase-Change Actuators for Physical Human–Robot Interactions. Supervisor: Amir Jafari. ” GPA: 4.00/4.00.

M.S., Biomedical Engineering, University of North Texas 2020–2021
Thesis: “Parkinson’s Disease and UPDRS-III Prediction Using Quiet Standing Data and Applied Machine Learning.”

B.S., Biomedical Engineering, University of North Texas 2017–2020
Minors: Mathematics, Mechanical Engineering.
Senior Capstone: “Hercules: Low Cost Trans-Humeral Prosthetic Capable of Power and Precision Grip Controlled Through Foot and Upper-Body Movement”

Appointments

Postdoctoral Associate, Material Robotics Laboratory, Boston University, Boston, MA 2026–Present

Soft optical shape sensing for steerable medical robots

Supervisor: Sheila Russo

Postdoctoral Researcher, Soft Biorobotics Perception Research Line, Istituto Italiano di Tecnologia (IIT), Genova 2024–2026

Monolithic soft robotics; integrated actuation, sensing & fabrication.

Supervisor: Lucia Beccai

PhD Researcher, Advanced Robotic Manipulators Lab, University of North Texas 2021–2024

Thermo-active variable impedance actuators; soft robotic systems.

Supervisor: Amir Jafari

Lab Affiliated, Biomedical AI Lab, University of North Texas 2020–2023

Biomedical machine learning; wearable sensor data.

Supervisor: Mark V. Albert

Peer-Reviewed Publications

(* = equal contribution; † = co-corresponding author)

Submitted / Under Revision

- [J.14] A. Coutinho*, **T. Exley***[†], P. Trunin, A. B. Nardin, L. Beccai[†]. Hollow-strut lattice architectures for distributed actuation and sensing in monolithic soft robots. *npj Robotics* (2026) (*submitted*)
- [C.8] **T. Exley**, A. Coutinho, L. Beccai. Selective Unit-Cell Actuation in Lattice Structures for Distributed Morphology in Soft Robots. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* (2026) (*accepted*)
- [C.7] P. Trunin, C. Gay, A. B. Nardin, **T. Exley**, D. Cafiso, L. Beccai. Bidirectional Optical Sensors for Actuation Tracking (BOAT) in Soft Lattice Systems. *2026 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)* (*accepted*)
- [B.1] D. Cafiso, **T. Exley**, P. Trunin, A. B. Nardin, S. Hajaresab Ladakhana, L. Beccai. Monolithic Strategies for Soft Robotics via Light-Induced 3D printing: Towards Fully Integrated Systems. Chapter in *Digital Light Processing and Stereolithography Additive Manufacturing*. (*submitted*)

In Press / Published

- [J.13] D. Johnson, **T. Exley**, M.-A. Torres, J. Slayton, M. Ecker, A. Jafari. Tailoring Silicone Mixtures for Soft Robotics: Predictive Modeling and Experimental Validation in Pneumatic Soft Actuators. *Bioinspiration & Biomimetics* (2026)
- [C.6] **T. Exley**, A. B. Nardin, P. Trunin, D. Cafiso, L. Beccai. Monolithic Units: Actuation, Sensing, and Simulation for Integrated Soft Robot Design. *IEEE RoboSoft* (2026)
- [J.12] P. Trunin, D. Cafiso, A. B. Nardin, **T. Exley**[†], L. Beccai[†]. MELEGROS: Monolithic Elephant-Inspired Gripper with Optical Sensors. *Advanced Science* (2026). [**Featured on Back Cover**]
- [J.11] **T. Exley**, D. Johnson, A. Jafari. Estimating Stiffness and Damping of a Novel Variable Impedance Actuator based on Adjusting Viscoelastic Properties of Thermoresponsive Polycaprolactone in Harmonic Motions. *Scientific Reports*. (2025)
- [J.10] **T. Exley**, R. Wijesundara, S. Wang, A. Moridani, T. Nilforooshan, A. Jafari. TVIM: Thermoactive Variable Impedance Module Evaluating Shear-Mode Capabilities of Polycaprolactone. *IEEE Access* (2025).
- [C.5] J. Jenkins, O. Madera, C. Guerrero, K. Humes, A. Malmquist, C. Renfrew, Q. Bakker, K. Siu, **T. Exley**, A. Jafari. Design and Implementation of a High-Precision Motor Control System for Adjustable-Stiffness Biomedical Treadmills. In *Proc. IEEE 18th Dallas Circuits and Systems Conf. (DCAS)*, 1–6 (2025).
- [C.4] F. Tajomi, **T. Exley**, A. Jafari. Development of a Miniature Thermal-Based Variable Impedance Actuator Using Thermoplastic Polymers for Scalable and Compact Robotic Applications. In *Proc. IEEE 18th Dallas Circuits and Systems Conf. (DCAS)*, 1–6 (2025).
- [C.3] **T. Exley**, R. Wijesundara, N. Tan, A. Sunkara, X. He, S. Wang, B. Chan, A. Jain, L. Espinosa, A. Jafari. Agonist–Antagonist Pouch Motors: Bidirectional Soft Actuators Enhanced by Thermally Responsive Peltier Elements. In *IEEE/RSJ IROS* (2024).

- [C.2] K. Ruiz, S. Ryan, S. Stutsman, T. Tirumala, J. Williams, R. Wijesundara, **T. Exley**, A. Jafari. Design considerations of Peltier-Integrated Therapeutic Wrist Wrap for Medical Applications. *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)* (2024)
- [J.9] **T. Exley**, D. Johnson, A. Jafari. Comparative Analysis of Peltier Devices and Flexible Heater Strips for Enhancing Bandwidth in Thermo-Active Soft Actuators. *Journal of Medical Robotics Research* 9(03n04):2450002 (2024).
- [J.8] **T. Exley**, E. Hays, D. Johnson, A. Moridani, R. Motati, A. Jafari. Toward a Unified Naming Scheme for Thermo-Active Soft Actuators: A Review of Materials, Working Principles, and Applications. *Robotics Reports* 2(1):15–28 (2024).
- [J.7] **T. Exley**, D. Johnson, A. Jafari. A Novel Variable Impedance Actuator Utilizing Adjustable Viscoelastic Properties of Thermoresponsive Polycaprolactone. *Robotics Reports* 1(1):57–66 (2023).
- [J.6] **T. Exley**, D. Johnson, A. Jafari. Towards a Novel Thermal-Based Variable Impedance Module through Adjusting Viscoelastic Properties of a Thermoresponsive Polymer. *IEEE Transactions on Medical Robotics and Bionics*, 1–1 (2023).
- [J.5] **T. Exley**, D. Johnson, A. Jafari. Utilizing the Peltier Effect for Actuation of Thermo-Active Soft Robots. *Smart Materials and Structures* (2023).
- [J.4] E. Hays, J. Slayton, G. Tejeda-Godinez, E. Carney, K. Cruz, **T. Exley**, A. Jafari. A Review of Rehabilitative and Assistive Technologies for Upper-Body Exoskeletal Devices. *Actuators* 12 (Soft Robotics in Biomedical Application):178 (2023).
- [C.1] Z. Liu, **T. Exley**, A. Meek, R. Yang, H. Zhao, M. V. Albert. Predicting GPU Performance and System Parameter Configuration Using Machine Learning. In *IEEE ISVLSI*, 253–258 (2022).
- [J.3] **T. Exley**, S. Moudy, R. M. Patterson, J. Kim, M. V. Albert. Predicting UPDRS Motor Symptoms in Individuals with Parkinson’s Disease from Force Plates Using Machine Learning. *IEEE Journal of Biomedical and Health Informatics* 26(7):3486–3494 (2022).
- [J.2] **T. Exley**, A. Jafari. Increasing Robustness and Output Performance of Variable Stiffness Actuators in Periodic Motions. *Mechanism and Machine Theory* 169:104645 (2022).
- [J.1] **T. Exley**, A. Jafari. Maximizing Energy Efficiency of Variable Stiffness Actuators through an Interval-Based Optimization Framework. *Sensors and Actuators A: Physical* 332:113123 (2021).

Grants & Funding

NIH G-RISE (T32GM136501)

2021-2024

Amount: \$109,318 USD.

Patents

- [P.4] A. Jafari, **T. W. Exley**. **Thermoactivate modular soft actuator based on phase transition**. US Patent No. 12,384,024, issued Aug. 12, 2025.
- [P.3] R. Kaleigh, S. Ryan, S. Stutsman, T. Tirumala, J. Williams, R. Wijesundara, **T. Exley**, A. Jafari. **Peltier-integrated therapeutic wrap**. US Patent App. 18/649,757 (2024).

- [P.2] E. Hays, E. Carney, K. Cruz, J. Slayton, G. Tejeda-Godinez, **T. Exley**, A. Jafari. **Devices for biomechanical assistance including an exoskeleton and a robotic glove**. US Patent App. 18/649,708 (2024).
- [P.1] A. Jafari, **T. W. Exley**. **Thermal-based variable impedance actuator**. US Patent App. 18/418,008 (2024).

Software & Datasets

- [S.1] **T. Exley**, A. B. Nardin. **Cushion**: a Grasshopper plugin for soft-robotics simulation powered by SOFA; developed from [C.6]. doi:10.5281/zenodo.21625292.

Awards

- [4] Outstanding Ph.D. Graduate, University of North Texas (2024)
- [3] Outstanding M.S. Student, University of North Texas (2021)
- [2] Outstanding Senior; Distinguished Honors College Scholar; UNT Excellence Scholarship (2020)
- [1] Esports Scholarship, UNT Varsity League of Legends Team (2019–2020)

Workshops

- [W.1] **T. Exley**, M. Taghavi, F. Vanneste, L. Beccai (organizers). "Monolithic Soft Systems: Co-design components, tools, materials, and fabrication strategies for next-generation robotics." *IEEE RAS RoboSoft 2026 Workshop* (full day), Kanazawa, Japan.

Projects

Deliverables, presentations, multi-partner coordination

PROBOSCIS (PROBOSCIdean Sensitive Soft Robot for Versatile Gripping)

European Union Horizon Grant, Agreement No. 863212.

Teaching

Traditional Courses

- [1] Teaching Fellow, UNT BMEN 2320 – *Biomedical Instrumentation* Summer 2021
- [2] Teaching Assistant, UNT BMEN 2210 – *Biomedical Data Acquisition*; BMEN 2320 – *Biomedical Instrumentation* 2020–2021

External courses, tutorials, and guest lectures

- [3] **Soft Robotics in i4.0**, Robotics in Industry 4.0: Challenges and Opportunities
University of Texas at San Antonio, Summer 2022
- [4] **Introduction to Machine Learning**
UNT Health Science Center, Jun 11, 2021

Research Supervised

Mentored >14 students individually as primary supervisor (now at **Harvard University, UC Berkeley, UT Austin, Georgia Tech, Rice University, Texas A&M**, University of North Texas, University of St. Thomas), plus 3 senior capstone groups at University of North Texas. Mentored students co-authored peer-reviewed conference papers [C.3], [C.5] and journal articles [J.8], and contributed to patent applications.

Service

Reviewer

Actuators; Applied Sciences; Biomimetics; IEEE Access; IEEE BioRob; IEEE ICORR; IEEE/RSJ IROS; IEEE RoboSoft; IEEE RA-L; IEEE T-MRB; IEEE T-RO; JNER; Robotics Reports; Soft Robotics

Media Coverage

- [2] H. Noel. *UNT Soft Robotics Lab Developing Patent-Pending Inventions*. University of North Texas Research & Innovation News, Apr 25, 2024. (*Featured subject; profiles two patents.*)
- [1] J. DeLeón. *Engineering Better Health*. The North Texan (UNT alumni magazine), Fall 2021. (*Profiled.*)

Professional Memberships

Institute of Electrical and Electronics Engineers (IEEE)	2020–Present
Biomedical Engineering Society (BMES)	2020–2024
American Society of Biomechanics (ASB)	2020–2021

References

Lucia Beccai, Ph.D., Senior Researcher, Istituto Italiano di Tecnologia. lucia.beccai@iit.it
Amir Jafari, Ph.D., Associate Professor, University of North Texas. amir.jafari@unt.edu
Vijay Vaidyanathan, Ph.D., Founding Chair of Biomedical Engineering, University of North Texas. Vijay.Vaidyanathan@unt.edu